LOGARITHMS: TEST PAPER

Topics: Evaluation, Properties, Equations, Inequalities

Name: _____ Date: _____ Total Questions: 61

Instructions: Attempt all questions. Each answer is shown in a box beneath its question.

SECTION A: EVALUATING LOGARITHMS

Q1. $\log_3 81$

ANSWER: 4

Q2. $\log_{27} 81$

ANSWER: 4/3

Q3. $\log_8 32$

ANSWER: 5/3

Q4. $\log_{27} 9\sqrt{3}$

ANSWER: 5/6

Q5. $\log_{1/3} (1/9)$

ANSWER: 2

Q6. $\log_{1/11} 1331$

ANSWER: −3

Q7. $\log_{2\sqrt{3}} 1728$

ANSWER: 6

Q8. $\log_2 \cos 45°$

ANSWER: −1/2

Q9. $\log_{2-\sqrt{3}} (2+\sqrt{3})$

ANSWER: −1

Q10. $\log_{0.125} 8$

ANSWER: −1

Q11. $\log_{3/4}$ 1.333... (recurring 1.333... = 4/3)

ANSWER: −1

Q12. $\log_{2.25}$ 0.444... (recurring 0.444... = 4/9)

ANSWER: −1

Q13. $\log_{10}$ 0.999... (recurring 0.999... = 1)

ANSWER: 0

Q14. $\log_{7\text{^}(1/3)}$ 2401 (base is the cube root of 7)

ANSWER: 12

Q15. $\log_2 \log_2 \log_3 \log_3 27^3$

ANSWER: 0

Q16. $\log_5 \sqrt{(5\sqrt{(5\sqrt{(5\ldots\infty)})})}$

ANSWER: 1 (the nested radical equals 5)

SECTION B: PROPERTIES OF LOGARITHMS

Q17. $\log_a b \times \log_b a$

ANSWER: 1

Q18. $\log_a b \cdot \log_b c \cdot \log_c d \cdot \log_d e \cdot \log_e a$

ANSWER: 1

Q19. $\log_2 3 \cdot \log_3 4 \cdot \log_4 5 \ldots \log_n (n+1) = 10$. Find n.

ANSWER: $n = 2^{10} - 1 = 1023$

Q20. $A = \log_{11} 11^{(\log_{11} 1331)}$

$B = \log_{385} 5 + \log_{385} 7 + \log_{385} 11$

$C = \log_4 (\log_2 (\log_5 625))$

$D = 10^{\log_{100} 16}$

Find AB / CD.

ANSWER: A = 3, B = 1, C = 1/2, D = 4
AB / CD = 3 / 2

Q21. If $N = \log_3 12 / \log_{36} 3 - \log_3 4 / \log_{108} 3$, then N is:
(i) composite (ii) prime (iii) co-prime with 7 (iv) rational but not an integer

ANSWER: N = 2
So N is prime (ii) and also co-prime with 7 (iii).

SECTION C: LOGARITHMIC EQUATIONS

Q22. Find x: $x^2 + 7^{\log_7 x} - 2 = 0$

ANSWER: x = 1 (x = −2 rejected since x > 0)

Q23. Sum of all solutions of $\log_{10} x + \log_{10} (x+2) - \log_{10} (5x+4) = 0$

ANSWER: Sum = 4 (x = 4 valid, x = −1 rejected)

Q24. Find x: $3^{(\log_3 x)^2} + x^{\log_3 x} = 162$

ANSWER: x = 9 and x = 1/9

Q25. Find x: $\log_{10} (2^x + x - 41) = x(1 - \log_{10} 5)$

ANSWER: x = 41

Q26. Product of all solutions of $x^{1 + \log_{10} x} = 100000x$

ANSWER: Product = 1 ($x = 10^{\sqrt{5}}$ and $10^{-\sqrt{5}}$)

Q27. Find x: $(x+1)^{\log_{10} (x+1)} = 100(x+1)$

ANSWER: x = 99 or x = −9/10

Note: my notes kept only x = 99 (using x > 0), but the real condition is x + 1 > 0. For x = −0.9: $(0.1)^{-1} = 10 = 100 \times 0.1$, so it works.

Q28. Find x: $\log_5 (5^{1/x} + 125) = \log_5 6 + 1 + 1/(2x)$

ANSWER: x = 1/4 and x = 1/2

Q29. Solve for x: $\log_2 (9 - 2^x) / (x - 3) = 1$

ANSWER: No solution (x = 3 makes the denominator 0)

Q30. Solve for x: $\log_{x-1} 4 = 1 + \log_2 (x-1)$

ANSWER: x = 3 and x = 5/4

Q31. Solve for x: $5^{1 + \log_4 x} + 5^{(\log_{1/4} x)^{-1}} = 26/5$

ANSWER: My notes stop part-way through this one.

Note: as written, it has no neat answer (numerically $t = \log_4 x \approx 0.024$ or -1.096). It is probably a typo in the question; please check the original.

Q32. Value of $[(\log_2 9)^2]^{1/\log_2 (\log_2 9)} \times (\sqrt{7})^{1/\log_4 7}$

ANSWER: 8

Q33. If $3^x = 4^{x-1}$, then x = ?

(A) $2\log_3 2 / (2\log_3 2 - 1)$

(B) $2 / (2 - \log_2 3)$

(C) $1 / (1 - \log_4 3)$

(D) $2\log_2 3 / (2\log_2 3 - 1)$

ANSWER: $x = \ln 4 / (\ln 4 - \ln 3) \approx 4.82$

A, B and C are all equal to this. D is not.

Note: my notes marked A as 'not possible', but $2\log_3 2 - 1 = \log_3(4/3) \neq 0$, so A is fine.

SECTION D: PRACTICE SET: SOLVE

Q34. (i) $\log_{x-1} 3 = 2$

ANSWER: x = 1 + √3

Q35. (ii) $\log_4 (2 \log_3 (1 + \log_2 (1 + 3\log_3 x))) = 1/2$

ANSWER: x = 3

Q36. (iii) $\log_3 (1 + \log_3 (2^x - 7)) = 1$

ANSWER: x = 4

Q37. (iv) $\log_3 (3^x - 8) = 2 - x$

ANSWER: x = 2

Q38. (v) $\log_2 (9 - 2^x) / (3 - x) = 1$

ANSWER: x = 0 (x = 3 rejected)

Q39. (vi) $\log_{5-x} (x^2 - 2x + 65) = 2$

ANSWER: x = −5

Q40. (vii) $\log_{10} 5 + \log_{10} (x+10) - 1 = \log_{10} (21x - 20) - \log_{10} (2x - 1)$

ANSWER: x = 3/2 or x = 10

Q41. (viii) $x^{1 + \log_{10} x} = 10x$

ANSWER: x = 10 or x = 1/10

Q42. (ix) $2(\log_x \sqrt{5})^2 - 3\log_x \sqrt{5} + 1 = 0$

ANSWER: x = √5 or x = 5

Q43. (x) $3 + 2\log_{x+1} 3 = 2\log_3 (x+1)$

ANSWER: x = 8 or x = 1/√3 − 1 ≈ −0.423

SECTION E: LOGARITHMIC INEQUALITIES (CONSTANT BASE)

Q44. $\log_{1/2} x < 3$

ANSWER: x > 1/8

Q45. $\log_3 (x-1) > 2$

ANSWER: x > 10

Q46. Domain of $f(x) = \log_{10}(x-1)$

ANSWER: (1, ∞)

Q47. Domain of $f(x) = \log_{2}(3-x)$

ANSWER: (−∞, 3)

Q48. $\log_{1/3}(5x-1) > 0$

ANSWER: (1/5, 2/5)

Q49. $\log_{0.5}(x^2 - 5x + 6) > -1$

ANSWER: (1, 2) ∪ (3, 4)

Q50. $\log_{1/4}(2-x) > \log_{1/4}(2/(x+1))$

ANSWER: (−1, 0) ∪ (1, 2)

Q51. $\log_{0.5}[\log_{6}((x^2+x)/(x+4))] < 0$

ANSWER: (−4, −3) ∪ (8, ∞)

Q52. Solution set of $2 - \log_{2}(x^2 + 3x) \geq 0$

ANSWER: [−4, −3) ∪ (0, 1]

Q53. If $\log_{1/3}((3x-1)/(x+2))$ is less than 1, x lies in:

ANSWER: (−∞, −2) ∪ (5/8, ∞)

Q54. $(\log_{2} x)^4 - [\log_{1/2}(x^5/4)]^2 - 20\log_{2} x + 148 < 0$

ANSWER: (1/16, 1/8) ∪ (8, 16)

Q55. If $\log_{0.3}(x-1) < \log_{0.09}(x-1)$, then x lies in the interval:

ANSWER: (2, ∞)

Note: my notes show some confusion here. Since $\log_{0.09} y = ½\log_{0.3} y$, the inequality becomes $\log_{0.3}(x-1) < 0$, so $x - 1 > 1$.

Q56. Solve $\log_{5/8}[2x^2 - x - 3/8] \geq 1$

ANSWER: [−1/2, −1/4) ∪ (3/4, 1]

SECTION F: LOGARITHMIC INEQUALITIES (VARIABLE BASE)

Q57. $\log_x(2x - 3/4) > 2$

ANSWER: (3/8, 1/2) ∪ (1, 3/2)

Q58. $\log_x(x^3 - x^2 - 2x) < 3$

ANSWER: (2, ∞)

Q59. $\log_{2x}(x^2 - 5x + 6) < 1$

ANSWER: (0, 1/2) ∪ (1, 2) ∪ (3, 6)

Q60. $\log_{x^2}(2 + x) < 1$

ANSWER: (−2, −1) ∪ (−1, 0) ∪ (0, 1) ∪ (2, ∞)

Q61. $\log_{(x+6)/3}[\log_2((x-1)/(x+2))] > 0$

ANSWER: $(-6, -5) \cup (-3, -2)$

Note: my notes give $(-5, -3) \cup (-3, -2)$. The slip is in Case 1 (base between 0 and 1): after flipping, $(x-1)/(x+2) < 2$ gives $(x+5)/(x+2) > 0$, i.e. $x < -5$ or $x > -2$, so Case 1 gives $(-6, -5)$. Case 2 gives $(-3, -2)$. Test $x = -4$: the value is negative, so it is not a solution.
